

Supplementary methods for The Index of Operations

S1. Reproducibility contract

Every result is identified by code version, dataset manifest, split, seed, and configuration. The fresh controlled benchmark is generated by `src/ioo_experiment.py`; it has no imports from the unrelated projects in the parent Downloads directory. A result is not promoted to the manuscript unless its JSON artifact exists and can be regenerated.

S2. Operator schema

An operator class stores a name, family, ports, units, and optional equations. An instance references one class and owns a variable subset. Interfaces store source, target, relation type, and parameters. This allows a motor instance to be replaced without changing the schema of the actuator it drives.

S3. Graph provenance

Physical edges are declared by the system description. Statistical edges are computed from telemetry and thresholded by absolute Pearson coupling. Temporal edges represent sequence adjacency. These relations are retained separately because a high correlation is not a causal proof.

S4. Controlled simulation

Each episode contains command, angular velocity, torque, winding temperature, housing temperature, and ambient temperature. Machine identity changes motor gain and thermal resistance. The substitution variant changes transmission ratio and cooling resistance. The recalibration transform is smooth and monotone channelwise, so it changes coordinates without changing rank order.

S5. Metrics

Balanced accuracy is used for identity retrieval because class counts can differ. Transfer degradation is the target score minus the source score. Recalibration stability is the absolute feature displacement before and after the warp. For every metric, report all seeds, mean, standard deviation, and the raw rows.

S6. Real-data extension

The first public-data targets are electric motor temperature telemetry and NASA C-MAPSS. The data must be downloaded through an authenticated Kaggle or local adapter, and splits must be

by profile or unit rather than by random windows. Any real-data paper table must include the dataset version, license, preprocessing, channel list, unit handling, missingness, and leakage audit.

S7. Cloud execution

Kaggle and Modal launchers are optional wrappers around the same experiment module. Google Colab runs the local module directly. Hugging Face publication is explicit and only uploads reviewed artifacts. Secrets are read from environment variables and never written to configuration files.

S8. Safety

The controller claims in this manuscript are representational and predictive. No learned policy is authorized to command physical hardware. A future control study requires a simulator, constraint monitor, fallback controller, human approval, and an independent safety review before hardware execution.